

Design Optimisation and Additive Manufacturing of an Aircraft Seat

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Abstract. This research work aims to provide an enhanced and optimized version of an aircraft passenger seat. The system can be likened to that of a commercial aircraft seat with the same seat leg, frame, armrest and similar values of operating parameters, and design considerations have been adopted, with the intent of mass reduction. Consequently, this can ensure lesser fuel consumption and curb emissions. Different aspects to designing the aircraft seat—such as ergonomics, manufacturability, environmental feasibility, etc are analysed. A basic aircraft seat adhering to the above has been modelled in a CAD software i.e, SolidWorks, which has then been exported to an advanced CAD software i.e, Fusion360, wherein Generative Design, an iterative computerized designing process in which the requirements of the user and the operating environment of the model are applied.. Moreover, topology optimization is carried out in nTopology, wherein material is directly removed via a FEA process. A significant mass reduction was observed. The aim of combining emerging technologies such as generative design and topology optimization along with additive manufacturing brings about the potential to compute optimized parts for rapid manufacturing solutions, as with the aircraft passenger seat.

INTRODUCTION

The aviation industry is an indispensable constituent for the global development of a country. It encompasses the transport of goods and people, and thus, the economy and societal value of that country. It paves the way for commerce, trade, tourism, etc. This priority leads to the question of the impact of the aviation industry on other deciding factors, such as the environment. The global fuel consumption by solely commercial airlines reached 95 billion gallons in 2019. This was before it plummeted to almost half the reported fuel consumption due to the COVID-19 pandemic [1]. Fuel economy is the underlying basis behind this project and fuel consumption, including its increasing costs, are still amongst the major concerns behind the aircraft industry. Furthermore, it is estimated to achieve net zero emissions by the global aviation industry by 2050 [2].

In order to provide a solution for this, numerical design optimisations, as suggested by Henderson et al., [3] can be undertaken to explore different parameters which can ultimately lead to lesser fuel consumption. Turgut et al. [4] also note the redundant fuel utilization that arises due to unnecessary mass considerations. One option is to minimize the overall weight of the aircraft. The average aircraft passenger seat weighs between 8 to 15 kgs. The aircraft seats alone contribute from 2240 to 4200 kg to the overall weight of the aircraft. Designs that could reduce the singular weight of one aircraft seat, without compromising on strength, could potentially decrease the net weight

by almost half. Mesquita [5] also suggests the need to choose materials capable of high mechanical strength, little brittle fracture, good corrosion resistance, low density to weight ratio, machinability and formability for sturdy structures. Previous studies by Putra and Sriwijaya [6] also mention requirement of safety factors above 1.33 and no plastic deformation upon application of test loads as a standard for aircraft certification.

Based on the criteria to incorporate all these factors, Generative Design was suggested as an optimisation tool. According to Plocher and Pansear (2019) [7], Generative Design merges the two schemes of topology optimization and latticing to provide a portfolio of multiple solutions, optimized with objectives for differing constraints. Aman [8] also investigated the validation of workflow of Generative Design and termed it as a tool that can revolutionize the future manufacturing industries.

A study conducted on the ergonomic analysis of the aircraft passenger seat outlined anthropometric data by Ozve Aminian and Izzuddin Romli [9] is referred to model a rudimentary aircraft seat on CAD software, i.e SolidWorks. Consequently, Generative Design, a tool implementing artificial intelligence algorithms on an iterative optimization basis to provide uncommon solutions that do not fit within conventional designs [10], is used. Through its computational methods, this elementary model of the aircraft seat can be honed into a low-weight, optimized version of the design that can subsequently maximize fuel efficiency. Depending upon the part of the aircraft seat to be optimized, the operational environment to be analyzed and the forces experienced by each face, edge or body during its performance are calculated so as to feed into the software, i.e Autodesk's Fusion 360, whose framework relies on cloud computing [10]. The geometry of the design to be demarcated into regions which were essential, called preserved bodies, and other regions such as the obstacle bodies could be manipulated by the program to provide multiple solutions. The software then provides the user with a number of possible outcomes that would best solve the problem as described. The different approaches to tackling the design with the aforementioned objectives in mind can be determined upon by choice of material, method of manufacturing and such. The final outcomes obtained from many iterations of the initial conditions fed into the software are then scrutinized and displayed towards the end, giving an impression of how a shape optimized version of an aircraft passenger seat could potentially look, while keeping the objectives for its design under purview.

In addition to Generative Design, design implementations were also computed using latticing for particular components for the seat, such as the armrest. Latticing converts solid components, such as the armrest of the aircraft seat, into stochastic arrangements of truss-based unit cells that can aid in the overall functionality of the part. For this project, the use of nTopology, can help create optimal lattice structures that could provide a solution for lightweighting in the armrest by selection of cell topologies, geometric parameters, etc.

The combination of Generative Design and latticing can put forth a myriad different designs of the aircraft seat, with varying characteristics, functionalities, etc. Bearing this in mind, the following analyses are aimed at working upon providing more feasible solutions to the problem statements of excessive fuel consumption by the aviation industry.

METHODOLOGY

The human computer interaction targeting the generative design process can be divided into three dynamic phases-Define; Generate and Explore [11]. The generative design process is done with the help of AI algorithms including convolutional neural networks, Generative adversarial networks [GANs] including Conditional GANs [CGANs], Conditional Wasserstein GANs [CWGANs], Boundary Equilibrium GANs [BEGANs] and so on [12]. Moreover, in Ivan Sosnovik's work, convolutional encoder-decoder architecture was familiarized and the issues with image segmentation tasks were stated [cite]. The problem to be addressed in this literature is weight reduction and an appropriate methodology has been proposed. When load is applied on a material, stress transmission is non uniform i.e, certain elements on the material are stressed less than the others. These volumes can be removed from the material, without compromising its strength and stiffness while achieving weight reduction. The key aspects that were taken into consideration beyond strength preservation and weight reduction are manufacturability, practicality of design so as to implement in real life scenarios and aesthetics.

Load Calculations

Essentially, the loads applied on the aircraft can be resolved into horizontal and vertical components. The vertical downward forces include the weight of an average passenger and the weight of the aircraft seat by itself. The weights of the passenger and aircraft seat are assumed to be 100 kg and 15 kg respectively. Maximum linear velocity

is taken to be 800 kmph by analyzing several aircraft speeds and the distance from the ground is taken as 2315ft. Moreover, the vertical and horizontal components of the total load varies while the aircraft is taking off, cruising and descending. With respect to the horizontal forces, the pseudo force experienced by the seat during acceleration is taken into account as specified by Noronha et al [13].

I. Vertical downward force

$$\begin{aligned} m_p &= \text{mass of person} = 100 \text{ kg} \\ m_s &= \text{mass of seat} = 15 \text{ kg} \\ m_t &= \text{Total mass on aircraft seat} = 115 \text{ kg} \\ F_t &= m_t g = 1128.15 \text{ N} \\ \text{Force on 1 leg} &= 564.075 \text{ N} \end{aligned}$$

When resolving the components for a take-off angle (α) of 15° and touch down angle (β) of 3° as shown in Fig. 1, the vertical load varies as follows.

$$F_{\alpha,v} = F_t \cos \alpha = 1090 \text{ N} \quad (1)$$

$$F_{\alpha,v} \text{ on 1 leg} = 545 \text{ N}$$

$$F_{\beta,v} = F_t \cos \beta = 1126.6 \text{ N} \quad (2)$$

$$F_{\beta,v} \text{ on 1 leg} = 563.3 \text{ N}$$

II. Horizontal backward force

Using the third kinematic equation of motion, with the maximum velocity obtained as 800 km/hr, the initial velocity of 0 km/hr and a ground roll distance of 706.6 m, the acceleration acting horizontally on the aircraft can be estimated as,

$$\begin{aligned} v^2 - u^2 &= 2as \\ a &= 34.91 \text{ m/s}^2 \approx 35 \text{ m/s}^2 \\ F_{pseudo} &= m_t a = 4025 \text{ N} \end{aligned}$$

$$\text{Pseudo force on 1 leg} = 2012.5 \text{ N}$$

When resolving the components for a take-off angle (α) of 15° and touch down angle (β) of 3° , the horizontal load varies as follows.

$$F_{\alpha,h} = F_t \sin \alpha = 292 \text{ N} \quad (3)$$

$$F_{\alpha,h} \text{ on 1 leg} = 146 \text{ N}$$

$$F_{\beta,h} = F_t \sin \beta = 60 \text{ N} \quad (4)$$

$$F_{\beta,h} \text{ on 1 leg} = 30 \text{ N}$$

However, while the aircraft descends, the $F_{\beta,h}$ The term in the equation (4) is taken to be negative, since the direction of load is opposite.

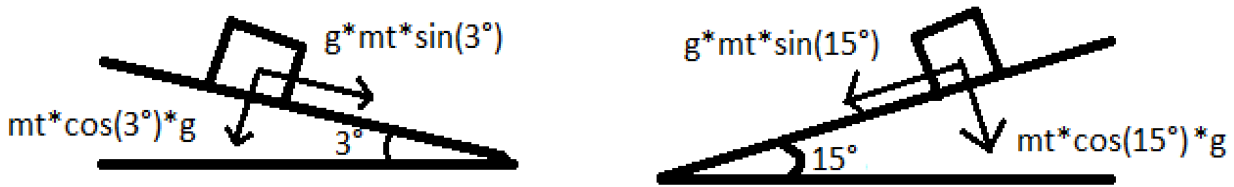


FIGURE 1. Diagram for load calculations

Topology Optimization of Seat Leg

Topology optimization is a computational design method having an aim to optimize the distribution of material in a given design space with respect to the loads and constraints, while maximizing the design's performance [14]. Following the calculation of the loads, a model of the quintessential aircraft seat was designed in Fusion360. It was imported to the topology optimization module in the same. A mesh with feature size 1 mm was created and loads were applied accordingly, following which the stress distribution was obtained in the simulation shown in Fig 2. The model was imported to the design module and then post processed.



FIGURE 2. Topology Optimization on Fusion 360

Post Processing

On performing topology optimization in Fusion 360 for the seat leg, the following simulation was set up as shown in Fig. 3. a). This mesh object was further post-processed using the same software through the Promote to Design Workspace option, indicated in Fi3. 3. b).

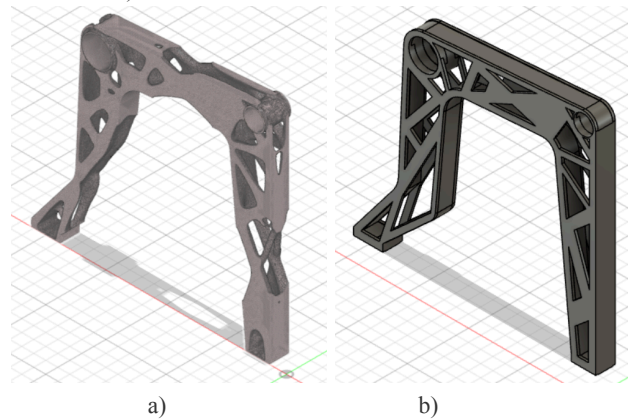


FIGURE 3. Post-Processing after topology optimization

Topology Optimization of Seat Leg in nTopology

Topology optimization in nTopology was carried out with a 50% mass reduction optimization constraint. Loads were applied as per the load calculations. The minimum feature size used for meshing is 2mm with linear geometric order. Material properties used for the FE model are of Al 6061-T6.

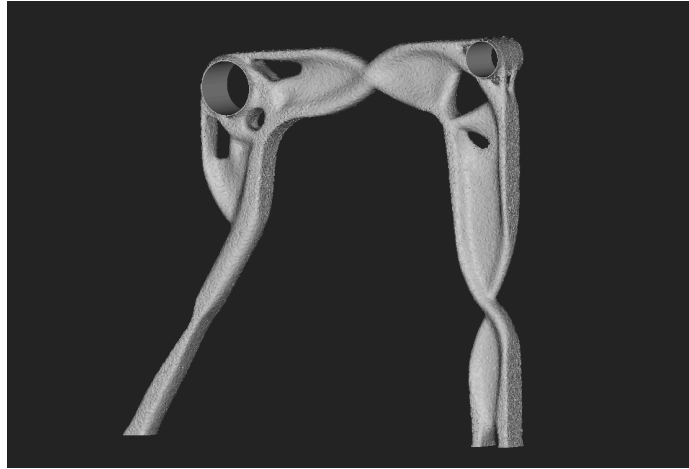


FIGURE 4. Topology Optimization on nTopology

The raw optimization result is shown in Fig. 4. The raw optimized part has several imperfections which then are rectified by smoothing the surface followed by other post processing activities in nTopology shown in Fig. 5.

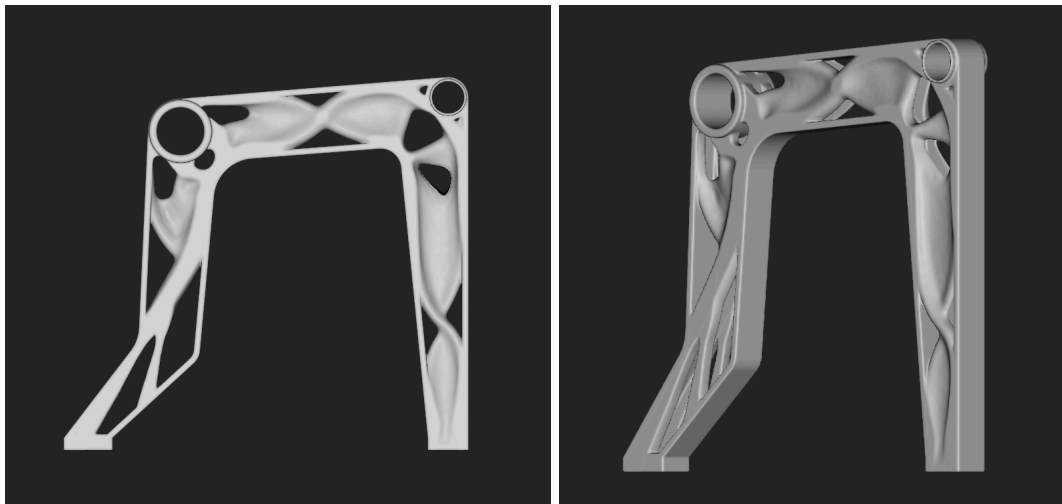


FIGURE 5. Post-Processing after topology optimization

The threshold for the element density field was set to 0.7, which can vary between 0, indicating no material, and 1 indicating solid material. To validate the design further model was exported as CAD file for simulation in ANSYS Static Structural. Although topology optimization is carried out so as to support the given loading conditions, a simulation was done so as to provide validation to our work done in nTopology, as shown in Fig. 6.

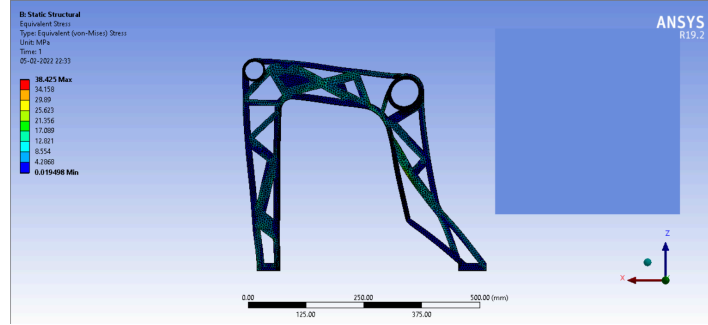


FIGURE 6. Validation of topology optimization

Generative Design

Generative Design is a new concept that harnesses the abilities of artificially driven technologies to provide high-performance design iterations that aim to solve complex discrepancies in anticipated designs such as factor of safety, mass reduction, increasing strength, etc. With respect to the aircraft seat, a standard model of a seat leg was imported with appropriate dimensions, properties and initial mass. The objective behind the study aims at weight reduction as well as material reduction. This means that the generative outcomes of the base model only deploy material wherein it is necessary or ‘preserved’. Remaining areas may or may not have material addition. The addition of ‘obstacle geometry’ ensures that parts of the model that must remain unchanged, are not tampered with. Whilst using generative design on a model, as observed by Pomazan and Sintea [15], it becomes imperative to define the design criteria by initiating boundary conditions in the algorithm. This includes manufacturing methods, tooling directions, supports for additive manufacturing, material employed, application of loads and constraints, objective of the outcomes, approximation in cost, etc., as well. Furthermore, as an example, Fenoon et al., clearly indicates the workflow and the loads required for different load cases with respect to the Generative Design of a mechanical pedal [16]. Thus, with the motive of generating a design outcome that would reduce the weight of the seat leg and maximize strength, a structural outcome study was set up as follows in Fig. 7 with the manufacturing methods selected for comparison being additive, 3-axis milling, 5-axis milling and unrestricted (a Fusion360 method where manufacturing constraints don't limit shape generation). The aluminum grades opted for are 6061-T6, 7075-T6 and Al A356 T6 based on versatility and AlSi10Mg for additive manufacturing (a Fusion360 default material selection). The explicit study outcome of a single component is shown in Table 1.

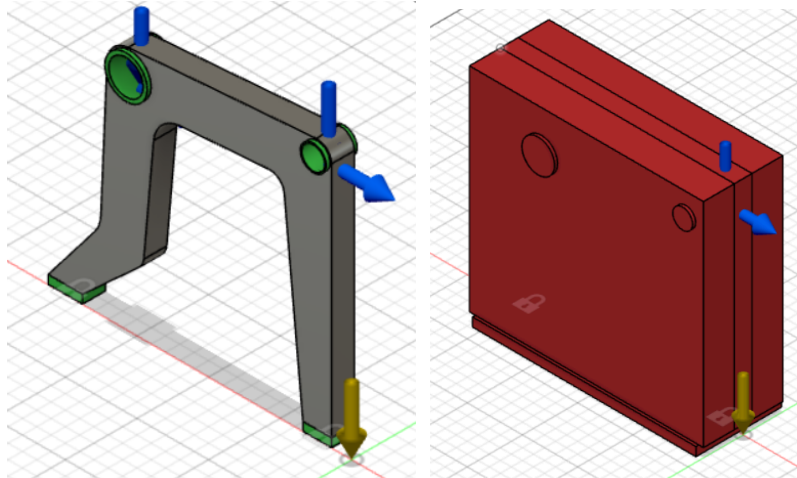


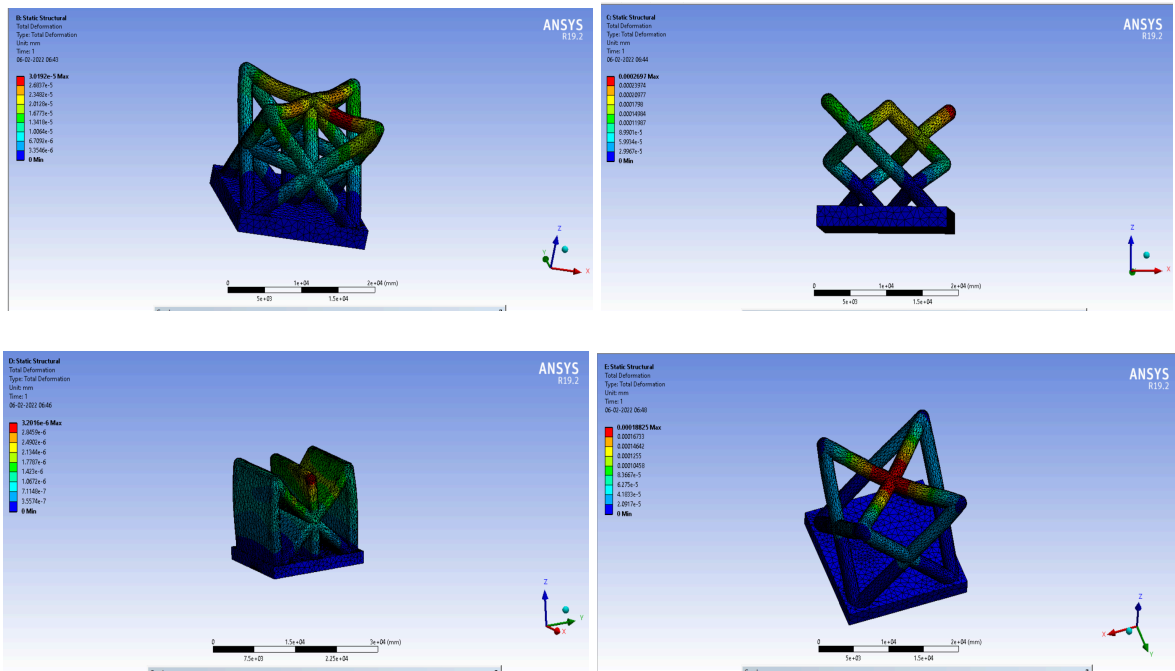
FIGURE 7. Study Setup.

TABLE 1. Structural study outcome

Structural Study Outcome 1	Completed
Properties	Completed
Status Material	Aluminum 6061
Orientation	X+, Y+, Z+, X-, Y-, Z-
Manufacturing method	3-axis milling
Visual similarity	Unique
Volume (mm ³)	1.214e+6
Mass (kg)	3.278
Max von Mises stress (Mpa)	5
Factor of safety limit	54.57
Min factor of safety	0
Max displacement global (mm)	

Latticing

Gibson and Ashby [17] defined the structures in which the cell walls have a common generator as two-dimensional cellular structures like honeycombs, and defined the structures in which cell walls have random orientations in space as three-dimensional cellular structures like foams. Lattice structures exhibit a multitude of applications, owing to their light-weighting abilities. The incorporation of these structures in the aerospace industry becomes imperative. The lattice structures as mentioned in the Table 2 were created in nTopology and the mesh file was imported to ANSYS Static Structural, wherein a comparative analysis of the structures were performed in order to determine the one with least deformation and high stress transmitting capability. Moreover, the mass of the components with these lattice structures were determined by simulation in nTopology. An encastre/fixed support was introduced in the base strust and a compressive load on 1000N was applied. Due to computational limitations, a mesh size of 1 mm was given, while the scaled dimensions of the lattice structures were 15mm x 15mm x 15mm. Hence, the deformation values are not accurate but are apt for comparison [18].



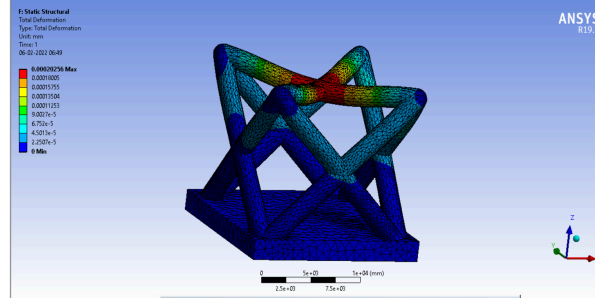


FIGURE 8. Lattice Structures

TABLE 2. Lattice Total deformation

Lattice Structure	Compressive Load (N)	Total Deformation (mm)
Octet	1000	3.1092e-5
Diamond	1000	26.97e-5
Honeycomb	1000	0.32016e-5
Body Centered Cube	1000	18.825e-5
Face Centered Cube	1000	20.256e-5

Armrest Optimization in nTopology

Optimization for lightweighting was also done for seat armrest. Load distribution data are taken from an experiment aimed to determine adults' seated body weight distribution in non-normal position [19]. The subjects were divided into two groups of male and female and a detailed description of their percentage body weight distribution on armrest i.e. 5.1% to 8.8% of body weight were taken as load values for simulation.

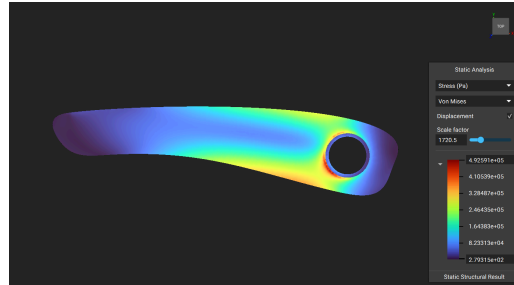
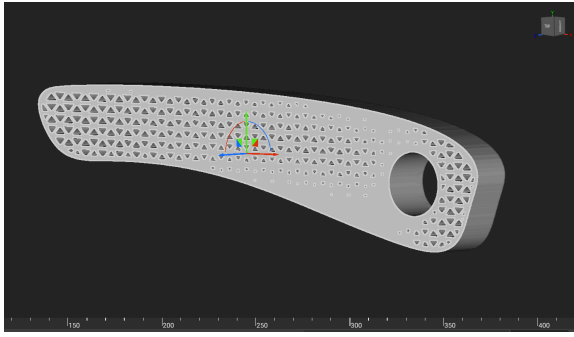


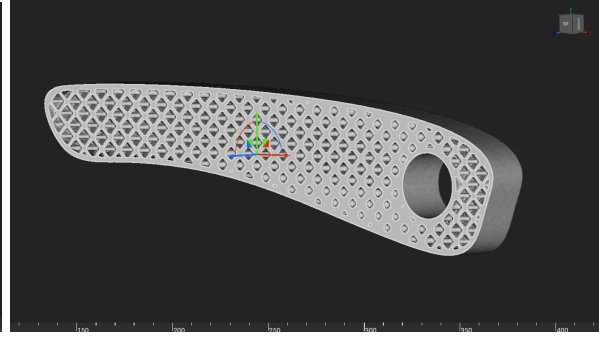
FIGURE 9. Static analysis of Armrest

Shell and lattice infill operation was carried out to remove the material and different types of lattices were used to compare between percentage weight reduction. Field driven design was used to capture the high stress area and vary lattice thickness from 2mm to 7mm from low to high stress region. Fig.9 shows the static analysis results and stress distribution across the armrest model. Stress values at different points are mapped to the thickness of lattice beams. Different lattice structures were used to compare between % weight reduction and total deformation values from the previous section.

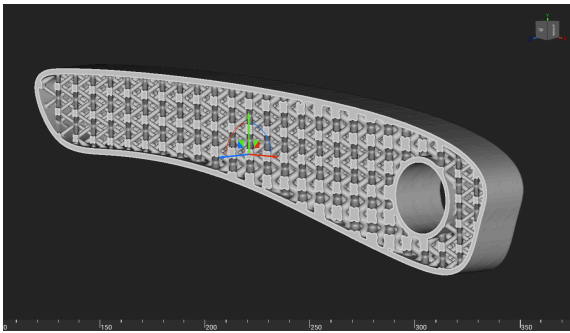
The cut section of the armrest below shows the detailed view of Shell and Lattice infill operation with field driven thickness based on stress values for lattice structures triangular honeycomb in Fig. 10.



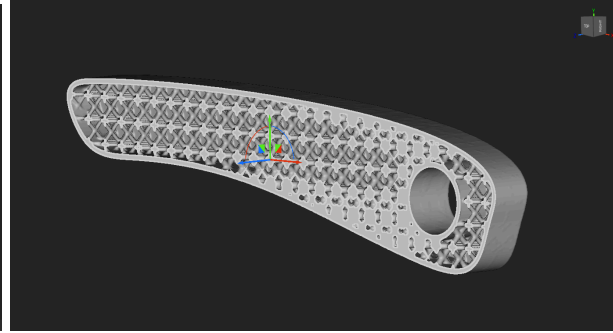
a) Triangular Honeycomb



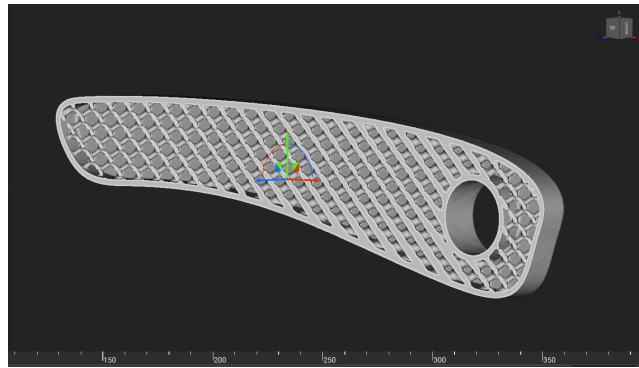
b) Octet



c) BCC



d) FCC



e) Diamond

FIGURE 10.

The original weight of the armrest was reduced to a significant level after Shell and Lattice infill operation. Table 3 shows the data for the reduced weight and percentage weight reduction from the initial value of 1.207 kg.

TABLE 3. Armrest mass and % weight reduction

Lattice Structure	Mass of Armrest (kg)	% weight change
Octet	0.843	30.1621
Diamond	0.633	47.5508
Triangular Honeycomb	1.072	11.1592
Body Centered Cube	0.632	47.5888
Face Centered Cube	0.672	44.3371

MATERIALS AND MANUFACTURING TECHNIQUES

A multitude of high stiffness and additively manufacturable materials are available, of which 5 such materials are recommended in this section, after an extensive literature study. These materials adhere to the rules set by the Federal Aviation Administration, USA.

Polyether ether ketone is a thermoplastic polymer and this material is chosen owing to its dimensional stability, broad chemical resistance and high melting point. The component can be directly incorporated in the final design by Fused Deposition Modelling (i.e, arm rest) or be used to create moulds, using which components (seat leg, seat frame) can be manufactured by conventional metal casting techniques, this combination is an emerging hybrid additive manufacturing technique. PEEK is a thermoplastic material and as such it can be converted into a range of component shapes and sizes by the full spectrum of fabrication technologies (extrusion, injection moulding, etc.) [20].

Al 6061 T6 is an aluminum alloy that has long been in use in the aerospace industry. However, as an off-eutectic with accompanying large freezing range, attempts to process the material by additive manufacturing have resulted in part cracking and diminished mechanical properties. Recently, a unique approach using mixed powders has been presented to process this historically difficult-to-process material. Expansion of this combined-powder approach to other materials systems not typically compatible with additive manufacturing is possible. Dense parts without solidification cracking have been produced by the SLM process, as verified using SEM and EDS [21].

AlSi10Mg is an aluminum alloy metal characterized with high strength or desirable mechanical properties and high thermal conductivity, primarily used in casting. The incorporation of magnesium in this alloy benefits in higher mechanical properties, increased modulus of elasticity and ductility[cite]. It is also extremely corrosion resistant. Furthermore, in addition to these properties, AlSi10Mg is present in the form of powder commonly used in additive manufacturing. Recent research [22] has come to show that AlSi10Mg is also suitable in selective laser melting (SLM), which is one of the powder bed based AM technologies. According to the sliced three-dimensional (3D) computer aided design (CAD) model, the metal products with complex freeform geometries are built by line-by-line and layer-by-layer laser scanning [23]. Either of these three additive manufacturing friendly materials can be used individually or in combination for aircraft seat parts.

Al 7075 is an aluminium alloy, primarily consisting of zinc, possessing high strength for components under high stress and strain. In a study by R. Clark Jr. et al.[24], several thermal treatments were developed which can encompass all the faulty heat treatments in aluminium alloy 7075-T6. This property makes it desirable for applications involving aircraft fittings, fuse parts, missile parts, aeronautical, aerospace and defense applications, etc. In comparison with Al 6061, Al 7075 is shown to be superior in terms of strength. However Al 6061 possesses greater weldability and is more corrosion resistant.

Al A356 is a casting alloy with very good machining properties. It has good weldability as well as excellent corrosion resistance. On heat treating it to give Al A356 T6, improved hardness is obtained. It can also be used in lieu of Al 6061. It is highly machinable as well and can be worked with milling, CNC machining, etc. On the other hand, A356 displays relatively poor mechanical characteristics due to coarse acicular eutectic silicon morphology, which is also to be taken into consideration [25].

TABLE 4. Materials and their properties [26].

Material	Density (g/cm ³)	Tensile Strength (MPa)	Yield Strength (MPa)	Young's Modulus (GPa)	Elongation at break (%)
AlSi10Mg	2.59	370-460	220-280	65-70	5-6
PEEK	1.26-1.32	70-110	87-96	3.76-3.95	50
Al 6061	2.7	310-330	276-290	68	12-17
Al 7075	2.81	572	503	71.7	9-11
Al A356 T6	2.67	228	152	73	3.5

RESULTS AND DISCUSSION

Topology Optimization of Seat Leg

The post processed Seat Leg was finally compared for final and initial mass properties. The initial weight of the model is 8.742 kg (approx) and after lightweighting there's significant weight reduction to 4.51 kg. The percentage weight reduction is obtained as 48.41 %. Cut section view of the optimized part in Fig. 11 a) and Original part in Fig. 11 b) is shown to compare the difference achieved between initial model and after final topology optimization model.

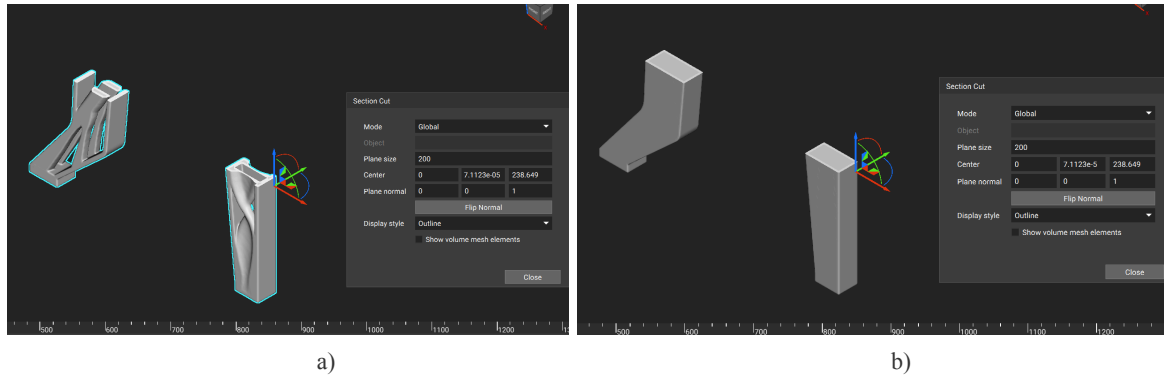


FIGURE 11.

Generative Design of Seat Leg - Comparative Analysis

Based on the workflow inputted to Generative Design, the outcomes were generated and completed. Those which converged are displayed in Table 5. The graph in Fig. 12 also shows the recommended outcomes, having a minimum factor of safety and reduced mass. From these, outcome 1 can be opted for as the final design in Fig. 13. Further post-processing can be done on this design.

TABLE 5. Comparative analysis of outcomes

Outcome Number	Material	Manufacturing Method	Mass Reduction (%)	Von Mises Stress (MPa)	Factor of Safety
1	Al-7075	Unrestricted	94.12	47.482	3.054
3	Al-7075	3 axis milling	94.19	30.425	4.766
4	Al-7075	5 axis milling	94.19	28.961	5.007
6	Al-6061	Unrestricted	94.35	39.335	7.957
8	Al-6061	3 axis milling	94.42	29.831	10.499
9	Al-6061	5 axis milling	94.43	30.447	10.28
11	Al-6061	Unrestricted	94.34	29.768	12.396
13	Al-6061	3 axis milling	94.42	29.07	12.694
14	Al-6061	5 axis milling	94.41	27.126	13.603
16	Al-6061	Unrestricted	94.14	29.232	15.291
18	Al-7175	3 axis milling	94.20	28.2	15.851
19	Al-7175	5 axis milling	94.21	30.591	14.612
21	Al-6061	Unrestricted	94.33	33.423	8.288
23	Al-6061	3 axis milling	94.41	27.693	9.93
24	Al-6061	5 axis milling	94.42	27.353	10.054
27	AlSi10Mg	Additive	94.32	32.183	7.457

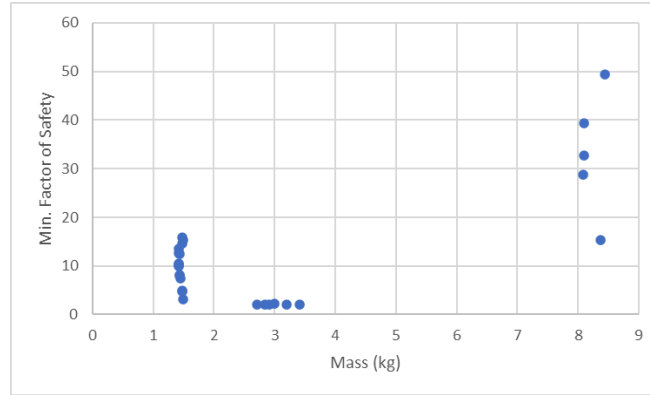


FIGURE 12. Recommended outcomes (converged, left side)



FIGURE 13. Final design is outcome 1

Latticing

A study for optimizing seat cushions of the seat was also done. The study was carried out with an objective to light weight seat cushions based on the occupant's body weight distribution on various seat cushions at different non-normal seated positions. Occupant weight data was taken [27] from two groups of male and female with different non-normal seated positions. The maximum subject load was taken as 98.3 kg and the measured load taken by both back and under thigh support seat cushion is taken as 76.45 ± 2.5 %.

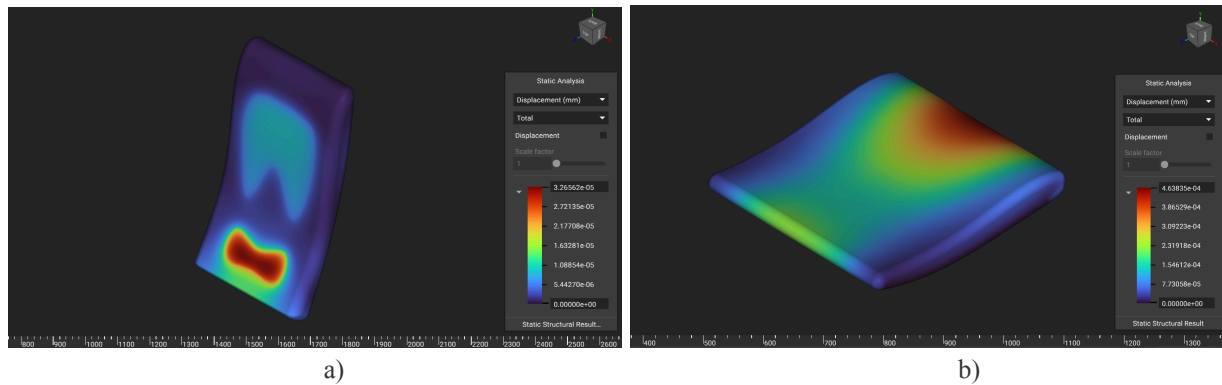


FIGURE 14. Static analysis of cushions

Fig. 14. Demonstrates the static analysis simulation for both Back seat cushion and Under thigh cushion. The material attribute taken is for silicon gel (SiO 2) [28]. The stress distribution data is then used as point map values for generating field driven lattice thickness for different periodic lattice bodies. Fig. 15.1 and Fig. 15.2 shows different lattice infill used for cushioning for Back Cushion and Under Thigh cushion respectively.

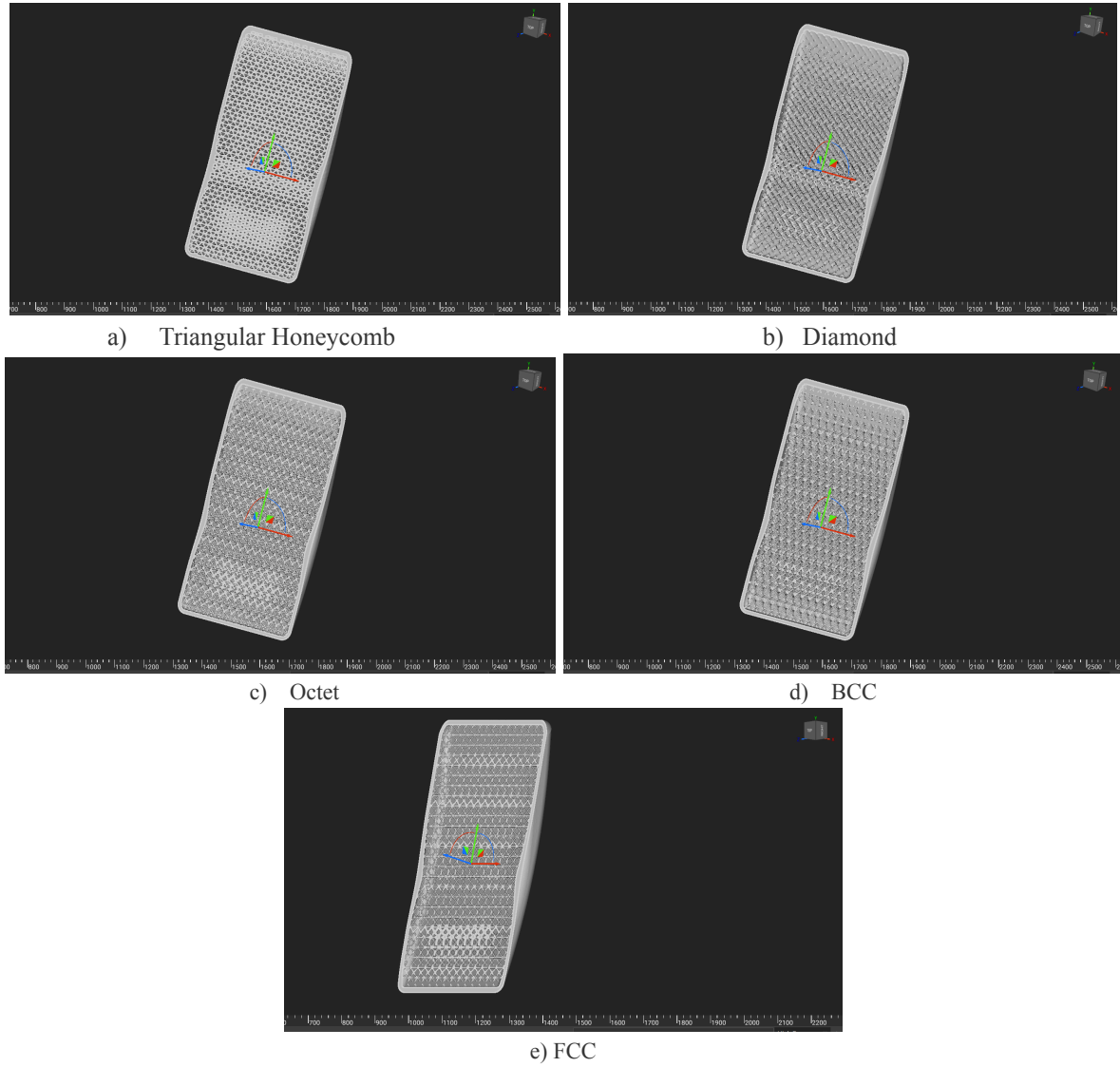


FIGURE 15.1

Table 6.1. shows the mass reduced from the initial value of 1.31 kg and the percentage mass reduction after the optimization of the Back seat cushion.

TABLE 6.1 Back seat cushion mass and % weight reduction

Lattice Structure	Mass of Back cushion (kg)	% weight change
Octet	0.640	51.343
Diamond	0.621	52.780
Triangular Honeycomb	1.003	26.653
Body Centered Cube	0.621	52.741
Face Centered Cube	0.663	46.570

Based on the data observations from Table 6.1. Back seat cushion with Diamond lattice is chosen as the final design.

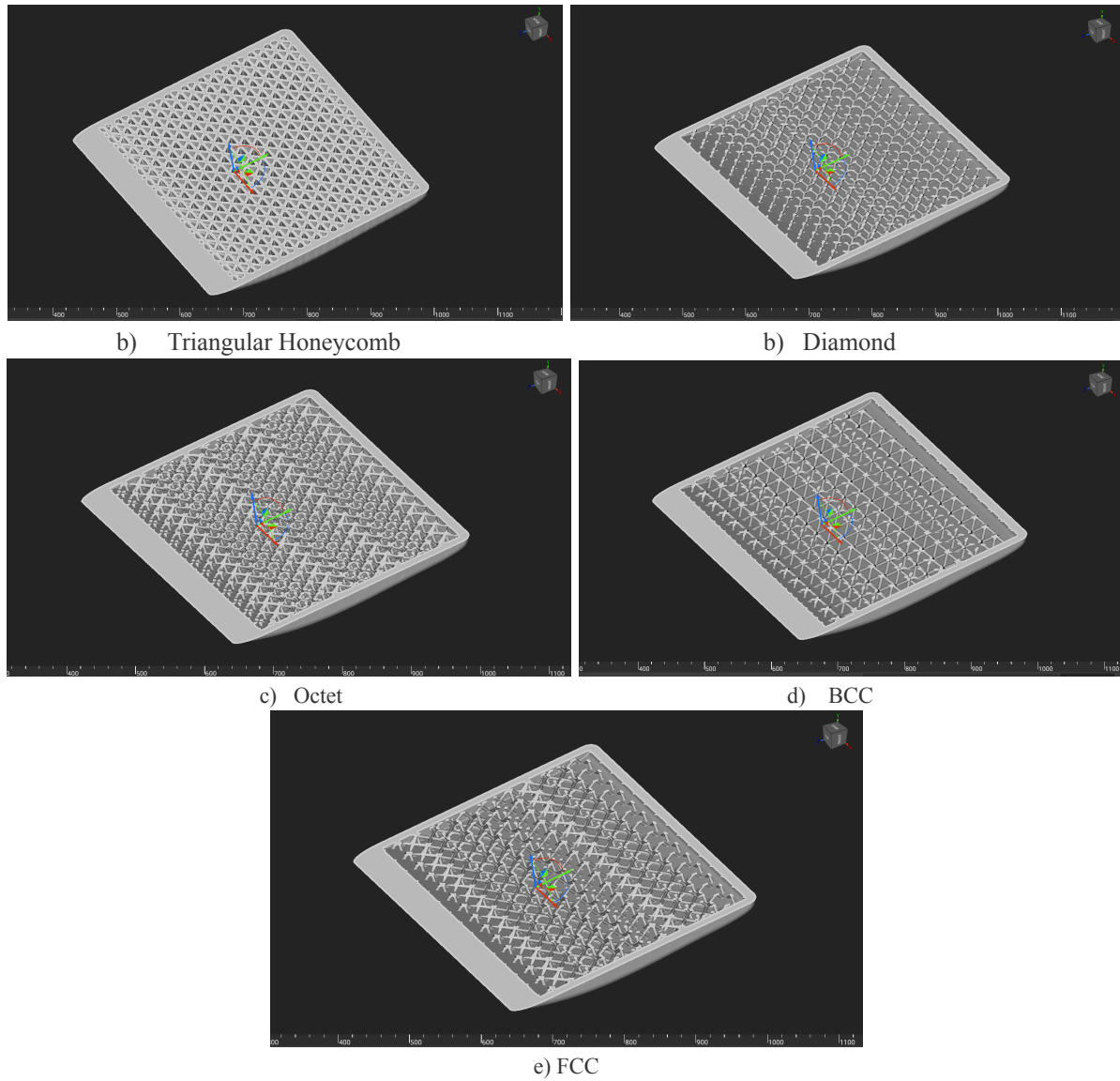


FIGURE 15.2

Table 6.2. shows the mass reduced from the initial value of 0.405 kg and the percentage mass reduction after the optimization of the Under thigh seat cushion.

TABLE 6.2 Under Thigh mass and % weight reduction

Lattice Structure	Mass of Under Thigh cushion (kg)	% weight change
Octet	0.287	29.1469
Diamond	0.267	34.1061
Triangular Honeycomb	0.341	15.6604
Body Centered Cube	0.267	34.0889
Face Centered Cube	0.271	33.0229

Based on the data observations from the above table Under Thigh seat cushion with Diamond lattice is chosen as the final design.

Mass reduction

Thus, through the numerical methods of topology optimization and Generative Design, the following components given in Table 7, indicate the mass reductions. Upon using Generative Design, the mass has reduced vastly with material additions being present only in necessary parts of the volume. As further scope, post processing can be implemented to this model.

TABLE 7. Reduction in masses

Component	Method Applied	Software Used	Initial mass (kg)	Final mass (kg)	Reduction in mass (%)
Seat leg	Generative Design	Fusion 360	25.419	1.494	94.11
Seat leg	Topology Optimization	Fusion 360	25.419	10.61	58.25
Seat leg	Topology Optimization	nTopology	8.742	4.510	48.41
Arm rest	Shell and Lattice infill	nTopology	1.207	0.632	47.58
Seat Cushion (Back)	Shell and Lattice infill	nTopology	1.31	0.621	52.78
Seat cushion (Under Thigh)	Shell and Lattice infill	nTopology	0.405	0.267	34.10

CONCLUSION

The goal of mass reduction was achieved and it is believed that incorporating these designs in a commercial aircraft would bring about a significant reduction in fuel consumed and emissions. The designs have come to be lightweighted, aesthetic and can be manufactured additively. As a continuation of thi research, the generatively designed seat leg can be post processed for real time use and the ergonomics factor of the chair can be scrutinised. The mass of art rest, seat cusion and leg were reduced and furthermore, using similar technology, side panels, housing mechanism and other parts of the aircrafts interior can be optimised as well.

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